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No Way to Sign Out? Unpacking Non-Compliance with Google Play's App Account Deletion Requirements

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Abstract

Despite the significant convenience mobile apps bring to our daily lives, the collection and use of personal information by these apps remain a major concern, particularly regarding how such data is handled after users sign out. To align with regulations like the General Data Protection Regulation (GDPR) that have included specific provisions granting individuals the right to request data deletion, mobile app stores, such as Google Play, have introduced new account deletion requirements that require apps to provide proper account deletion methods. In this work, we conducted the first study on investigating non-compliance issues with Google Play's app account deletion requirements. Starting with a pilot study of the top 50 apps on Google Play, we identified potential issues related to account deletion and defined three main categories of issues: link issues, content issues, and functionality issues. Based on these findings, we developed a tool named DELETETRACKER to automatically collect account deletion-related information from Google Play and semi-automatically identify non-compliance issues regarding account deletion. Using DELETETRACKER, we analyzed 863 Google Play apps' account deletion information. Among the 494 apps with accessible account deletion links, DELETETRACKER found only 8.5% of apps to provide both in-app path and web-based account deletion methods, which fully comply with Google Play's account deletion requirements. 64.6% of apps offer only one account deletion method. We also found 12 apps that failed to delete user accounts. We have reported our findings to Google through the vulnerability reporting process. Following our disclosure, Google acknowledged the reported issue and assigned it a Medium (S2) severity level.

1 Introduction

Google Play [9] is one of the leading mobile app platforms and offers more than two million apps as of 2025 [11]. These mobile apps may collect various types of user information, e.g., emails, phone numbers, and addresses, for the apps' func-

tionality. To better protect users' privacy, government agencies in different regions have published a list of legal and lawful regulations, e.g., the General Data Protection Regulation (GDPR) [7] and the California Consumer Privacy Act (CCPA) [5]. Marketplaces like Google Play also define their own policies on protecting user data and require app developers to disclose data practices to users, such as providing a privacy policy or providing privacy-related information in an app's Data Safety Section (DSS).

Many apps require users to sign up or register an account to use their features which involve storing user accounts and associated data in their databases or servers. Therefore, in regulations like GDPR, one specific policy is about users' right to protect their data, such as the right to erasure (delete) their data on an app [3]. For these apps, the right regarding personal data deletion ensures that when users sign out and request account deletion, their account and associated data should be permanently removed. In alignment with these regulations, Google Play recently introduced requirements for developers of apps published on Google Play to provide users with account deletion methods [23]. In Google Play DSS (Data Safety Section), developers are required to provide account deletion methods if their app allows users to create an account within the app. Specifically, developers must offer two ways for users to delete their accounts and associated data: an *in-app* path and a *web link resource* method. Starting in May 2023, Google Play introduced stricter enforcement measures, including restricting app updates and removing apps that did not offer the above account deletion methods.

A few works have examined the data deletion on the Web [35, 36, 43] and account deletion on social media [34, 47]. However, little is known about how Google Play's new account deletion requirements have been implemented in practice across apps on the platform. Specifically, it remains unclear whether app developers actually provide an account deletion link for their apps, whether these links comply with Google Play's requirements, and whether users are truly able to delete accounts following the mechanisms provided. Failing to provide proper account deletion methods can result

in non-compliance with Google Play’s app account deletion requirements and potential violations of privacy regulations. Moreover, it can undermine user trust in the platform and negatively impact the overall user experience. Therefore, it is critically important to detect account deletion issues in mobile apps currently available on the Google Play platform.

In this study, we aim to assess the extent of app developers’ compliance with Google Play’s recent account deletion requirements, which also provides insight into Google Play’s enforcement of the policy. Specifically, we aim to address the following research questions:

- **RQ1 (Availability):** Do apps provide accessible account deletion links?
- **RQ2 (Compliance):** Do the links offer the required account deletion methods to comply with Google Play’s app account deletion requirements?
- **RQ3 (Functionality):** Can users successfully delete their accounts following the methods provided by developers?

To answer the above research questions, we first conducted a pilot study and manually analyzed the top 50 Google Play apps to identify potential issues related to account deletion. Motivated by the preliminary findings, we designed a tool named DELETETRACKER to detect account deletion issues in large-scale Google Play apps. To the best of our knowledge, this is the *first* work to detect and analyze account deletion issues in the Google Play DSS. Our contributions are summarized as follows:

- **Potential issues identification through a pilot study:** To identify potential account deletion issues, we first conducted a pilot study and manually analyzed the top 50 apps on Google Play. Surprisingly, only 6 apps fully comply with Google Play’s app account deletion requirements, which mandate providing both an in-app path and a web link resource for account deletion. We identified various issues, including missing or inaccessible links, cross-platform inconsistencies, providing no/only one/other alternative account deletion methods, and the failure to delete accounts. This pilot study provides insights and guidance that enable us to formally define the problem of detecting account deletion issues within Google Play DSS.
- **New tool development:** Based on the findings from our pilot study, we developed a tool named DELETETRACKER to detect account deletion issues. The tool consists of four modules: Account Deletion Link Crawler crawls account deletion information from the Google Play mobile app and website; Account Deletion Link Analyzer detects issues related to the account deletion links such as missing, inaccessible, similar links, and cross-platform

inconsistency; Account Deletion Method Classifier identifies issues within the content of these account deletion links, including missing account deletion methods, other alternative account deletion methods, and missing in-app path or web-based account deletion methods; and Account Deletion Validator verifies whether an account is truly deleted by performing: registering, logging in, deleting the account, and attempting to re-login/re-register.

- **Large-scale analysis and issue disclosure:** Using DELETETRACKER, we successfully collected 863 apps with their account deletion information. Among these, 291 apps do not provide account deletion links, and 25 apps’ account deletion links are inaccessible. After that, we classified account deletion methods for 494 apps that provide an account deletion link. DELETETRACKER found that 13.8% of apps only offer an in-app path account deletion method, 50.8% provide only a web-based account deletion method, and just 8.5% comply fully with Google Play’s app account deletion requirements by offering both methods. 12 apps failed to submit account deletion request or failed to delete user accounts even after account deletion requests were submitted and the apps indicated that the account had been deleted. We have reported our findings to Google and shared the DELETETRACKER and findings to facilitate future research¹. We discuss the responsible disclosure in the Ethics Consideration section.

2 Background

2.1 Privacy-Related Lawful Regulations

To protect users and data privacy, a list of lawful regulations has been published in different regions. In 2016, the European Union (EU) passed the General Data Protection Regulation (GDPR) [7], widely regarded as the strictest privacy regulation in the world [26]. The GDPR consists of 11 Chapters with 99 Articles covering a wide range of data protection requirements. In Chapter 3, GDPR defines different rights of the data subject, including the right of access by the data subject, the right to rectification, the right to data portability, and the right to object. In particular, Article 17 of the GDPR defines the “Right to Erasure”, which grants individuals the right to request the erasure of personal data concerning them without undue delay, and controllers are obligated to comply with such requests promptly. Followed by GDPR, in the United States, the California Consumer Privacy Act (CCPA) [5], which was enacted in 2018, also provides consumers with the right to request the deletion of collected personal information. Moreover, it requires businesses to respond to such requests within

¹The details of the top 50 apps, our tool and findings are available at <https://zenodo.org/records/15610882>.

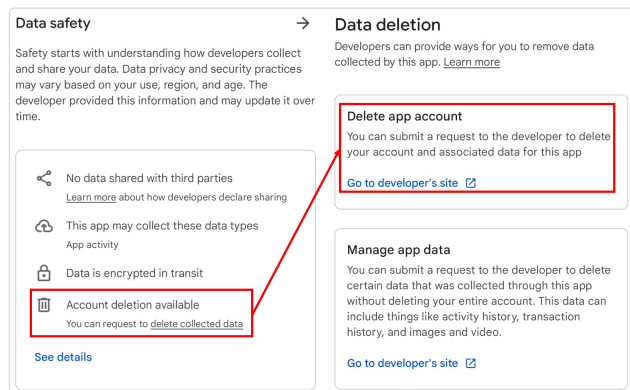


Figure 1: “Data safety” section (left) and “Data deletion” section (right) of a Google Play app. Users can click “Data Safety” to enter the “Data deletion” section.

45 calendar days. More recently, several U.S. states passed regulations with similar requirements [21], including the Virginia Consumer Data Protection Act (VCDPA) [25], the Colorado Privacy Act (CPA) [6], the Connecticut Data Privacy Act (CT-DPA) [20], and the Utah Consumer Privacy Act (UCPA) [24]. All of them require businesses to delete consumers’ personal data upon request.

2.2 Account Deletion Requirements of Google Play Platform

In 2022, Google introduced a new section named Data Safety Section (DSS) to disclose how developers collect, share, and protect user data [17]. Developers are required to submit DSS information through a Data Safety form and all developers with published apps on Google Play must complete it. The DSS includes several sections, such as data collected and data shared with third parties.

In April 2023, Google Play announced the new account deletion requirements and added a new data deletion section in DSS, as shown in Figure 1. After developers update app information and provide the account deletion link, users can click “delete collected data” to access the data deletion section details and then click the “Go to developer’s site” link to get more information about how to delete their account. Google Play has a clear requirement for account deletion: “If your app enables account creation, you must: provide users with an *in-app path* to delete their app accounts and associated data; and provide a *web link resource* where users can request app account deletion and associated data deletion.” Therefore, we will focus on the two methods for account deletion. Temporary measures such as account deactivation, disabling, or freezing do not meet Google’s definition of account deletion. However, Google claims that “if certain data must be retained for legitimate purposes, such as security, fraud prevention, or compliance with regulatory requirements, developers must

clearly communicate these data retention practices to users, typically through their privacy policy” [1].

Similar to Google Play, Apple launched its Apple Privacy Label (APL), which is similar to Google DSS, to help users understand the app’s privacy practices [22] in 2020. Since June 2022, Apple has also required apps with account creation functionality to provide users with an account deletion method within the app. However, Apple does not specify the exact format of account deletion or mandate that includes account deletion in the APL. Instead, Apple has outlined several guidelines and recommendations for account deletion [16]: 1) The account deletion option should be easy to locate within the app; 2) Account deletion must involve the removal of both the account and associated personal data; 3) Apps may also provide a website-based account deletion option, but the link must direct users to the account deletion page directly; 4) If account deletion cannot be completed immediately, users should be informed about any additional time required to finalize the process, etc. These requirements are similar to Google Play’s app account deletion requirements but leave more room for developers, making compliance checking more difficult. In this work, we only focus on the Google Play platform.

3 Pilot Study and Problem Definition

Since there is no existing work on analyzing the account deletion section of Google Play apps, we conducted a pilot study and manually tested the top 50 apps from Google Play to gain an understanding of the current account deletion section situation in Google Play DSS.

Manual exploration procedure: Two PhD-level students with prior research experience in account deletion (these two students served as the annotators throughout the study) acted as regular users to interact with Google Play apps using Android Studio [2] virtual machines. For each app, they first searched the app on the Google Play and then checked whether an account deletion link was provided in the app’s DSS. If a link was available, they clicked on it to assess its accessibility. For each accessible link, they carefully reviewed the content of the link (e.g., textual instructions, forms, buttons, and other relevant UI elements) and followed Google Play’s policy to identify the account deletion methods (i.e., in-app and web-based account deletion) or any potential non-compliance issues. The two annotators independently documented their findings and resolved any disagreements through discussion. For example, two annotators initially disagreed on app *Walmart: Shopping & Savings*. Annotator 1 considered that the app did not provide any account deletion method, as the account deletion link led to a technical error page. Annotator 2 opened the same link on a mobile device, which redirected to the account settings page, and located a “Delete Account” button after navigating through the app. Therefore, annotator 2 labeled it as providing an in-app path. After further discussion, both annotators agreed that the link neither

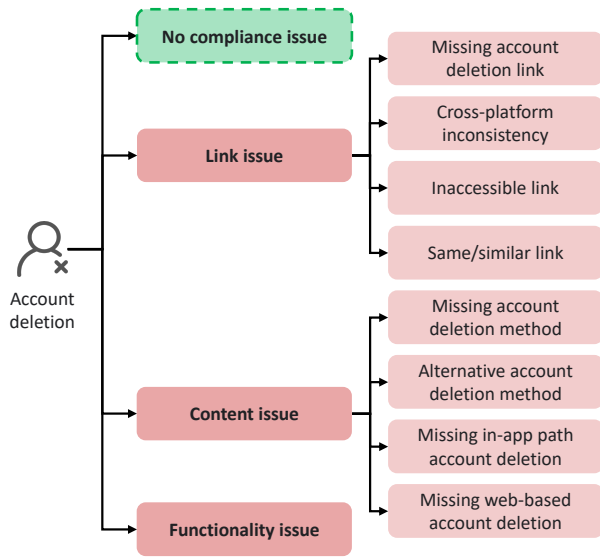


Figure 2: Problem definition of different issues in the account deletion process.

led directly to the account deletion interface nor provided a clear path for users to locate the deletion option. As a result, they classified the app as “*missing account deletion method*”.

Based on the issues identified, we classified the results into one compliance category and three types of non-compliance, as shown in Figure 2. Our classification criteria for non-compliance issues follow an outside-in approach: 1) **Link level**: whether the account deletion link is missing, inaccessible, or inconsistent between the app and the website; 2) **Content level**: whether one or both required deletion methods (in-app and web-based) are missing, or whether alternative methods are provided in the content of the account deletion link; 3) **Functionality level**: whether the provided account deletion method is non-functional. Based on our exploration results and classification criteria, we provide detailed textual definitions, screenshots, and app examples with findings for each category in the following subsections.

3.1 Compliance with Google Play’s App Account Deletion Requirements

Definition: An app must provide an account deletion link in the DSS. Within this link, the app needs to provide both in-app path and web-based account deletion methods. The in-app path account deletion method requires apps to offer a path like the step-by-step tutorial on how to delete an account within the app. The web-based account deletion method requires apps to provide a web link resource where users can request app account deletion.

Finding: We found 6 apps in our pilot study providing account deletion links with both in-app path and web-based account deletion methods. Typically, these apps offered step-

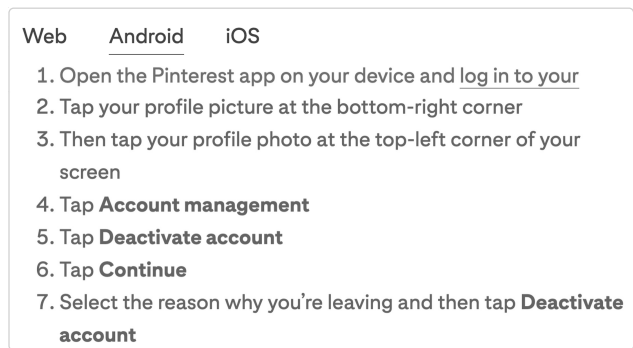


Figure 3: Account deletion page from *Pinterest* app.

by-step tutorials that contain several sections for each platform, such as for Android, IOS, and Web platforms, or allowed users to select their platform to view the corresponding account deletion tutorial. Figure 3 shows an example of *Pinterest*’s account deletion page. It provided three tutorials for different platforms, and users could delete their accounts from mobile apps or websites.

3.2 Account Deletion Link Issues

We identified some apps that did not provide an account deletion link when it was supposed to, or offered an inaccessible link. For link issues, we only focused on the account deletion link (URL), not the content it led to. Our method for automatically detecting link issues will be introduced in Section 4.2.

3.2.1 Missing Account Deletion Link

Definition: An app does not have a “Go to developer’s site” URL within the “Delete app account” section.

Finding: Ten apps in our pilot study did not provide account deletion links in the DSS. For instance, the *Beat Maker Pro - DJ Drum Pad*’s DSS indicated “Developer hasn’t provided information about data deletion”. The *Google Messages* app allowed users to register a Google account within the app but did not provide an account deletion link. Additionally, there were 6 apps (e.g., *PDF reader - PDF Viewer*, *QR & Barcode Scanner Plus* and *Remote Control for TV - All TV*) without traditional account registration or login features. Instead, they had a subscription function that required users’ sensitive information, such as contact information, billing addresses, and card numbers. These subscriptions could function as accounts since they link personal information to a user-specific record, and these apps might need to provide links for data deletion.

3.2.2 Cross-Platform Inconsistency

As shown in Figure 1, the “Data deletion” section in the Google Play mobile app may contain two subsections: “Delete app account” and “Manage app data.” If these two

subsections are absent, the app is considered not to provide data deletion functionality. The Google Play website provides a data deletion statement, “*You can request that data be deleted*,” to indicate the data deletion functionality for each app. We identified inconsistencies in the same app’s data deletion functionality between the Google Play mobile app and the Google Play website.

Definition: If the “Data deletion” section in a mobile app is empty (neither of the two links exists), but the website displays the data deletion statement, this is considered an inconsistency. In contrast, if the “Data deletion” section in the mobile app is not empty (at least one link exists), but the website does not display the data deletion statement, this is also considered an inconsistency.

Finding: Four apps in our pilot study had cross-platform inconsistency issue (i.e., *Google Messages*, *PDF Reader - PDF Viewer (QR Code Scanner)*, *PDF Reader - PDF Viewer (WeFindertools)*, and *All Video Downloader & Player*). Figure 4 shows the cross-platform inconsistency in the *All Video Downloader & Player* app.

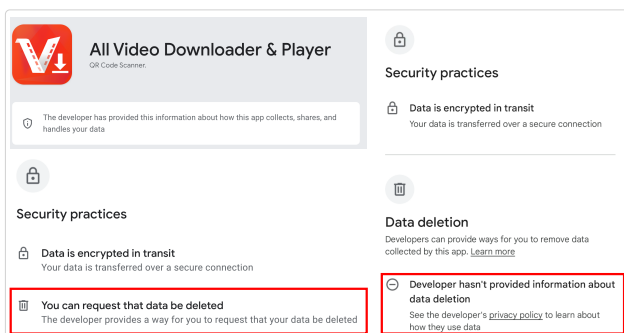


Figure 4: Inconsistent data deletion information on Google Play website (left) and Google Play mobile app (right) for the same app.

3.2.3 Inaccessible Links

Definition: An app provides the account deletion link in its DSS section, but the link is inaccessible.

Finding: No inaccessible link was found in the pilot study.

3.2.4 Same/Similar Links

Definition: Different apps share identical URLs or the same domain for their account deletion links.

Finding: For example, *YouTube TV: Live TV & more* and several other apps owned by Google share the identical account deletion link². The *Instagram* and *Thread* apps shared similar links with the same domain (i.e., *instagram.com*). Using the same or similar links could lead to further issues, which will be discussed in Section 3.4.

²<https://support.google.com/accounts/answer/32046>

3.3 Account Deletion Content Issues

We identified 34 apps with content issues. These apps provided an accessible account deletion link; however, the content within the link failed to comply with Google Play’s app account deletion requirements. Section 4.3 describes our tool to detect these issues.

3.3.1 Missing Account Deletion Method

Definition: The content within the provided account deletion link does not contain any information guiding users on how to delete their accounts.

Finding: We found 8 apps’ account deletion links did not provide an account deletion method. Among them, *Walmart* and *Cash App*, redirected users to pages that do not provide any tutorial on how to delete their accounts. For *Walmart* app, when users clicked the account deletion link on a mobile device, it opened the Walmart app directly instead of navigating to an account deletion page, leaving users unable to find a deletion path through the link. When we copied and pasted the link into the computer browser, there was still no useful information about account deletion. For *Cash App*, the link redirected to a login page but failed to send the required code to the input phone number, preventing users from logging in and accessing potential account deletion information.

Among the 8 apps, 3 apps (i.e., *Spotify: Music and Podcasts*, *CapCut - Video Editor* and *CVS/pharmacy*) did not include any useful information about account deletion method within their provided link. However, they provided the account deletion method on other web pages, where the related information could only be found by manual search. For example, in the *Spotify: Music and Podcasts* app, the provided link directed users to an article titled “Data rights and privacy choices”, which contained nine collapsible sections without visible information about account deletion. After expanding all sections and searching for keywords such as “delete”, “close”, or “cancel”, the only related information was the statement, “To delete that personal data, you need to close your account”, still not explaining how to close the account. However, if users entered “delete account” in the search bar above the article, they would be redirected to the correct page for account deletion. We did not consider this an adequate account deletion method, as it is very difficult to find the correct process without extensive effort and precise keyword searches.

3.3.2 Alternative Account Deletion Method

Definition: An app only offers alternative account deletion methods, e.g., sending an email, in the app’s account deletion links, which are not listed in Google Play’s app account deletion requirements (i.e., in-app path or web-based account deletion method).

Finding: Three apps (i.e., *Instabridge: WiFi Map*, *Health Sense: Blood Sugar Hub* and *ShortMax - Watch Dramas &*

Show) required users to send an email to the app developers to delete their account. We consider only providing such alternative methods to be an issue since Google Play’s app account deletion requirements specify only two account deletion methods (in-app path and web-based method).

3.3.3 Missing Web-based Account Deletion Method

Definition: An app only provides an in-app path account deletion method but fails to provide a web-based account deletion method.

Finding: Six apps only provided an in-app path account deletion method. For example, the McDonald’s account deletion link included an in-app path as follows: *launch the McDonald’s app → select “More” on the bottom navigation bar → select the “Profile” topic → select “Delete account”*. However, it did not mention how to access a webpage for account deletion or provide a direct link to the web-based account deletion method.

3.3.4 Missing In-app Path Account Deletion Method

Definition: An app only provides a web-based account deletion method but does not provide an in-app path account deletion method.

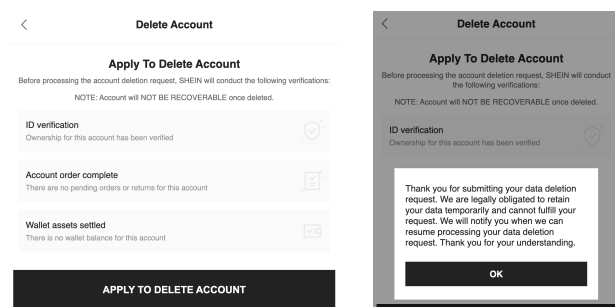
Finding: Seventeen apps only provided a web-based account deletion method, and there were mainly three sub-types with web-based account deletion methods: 1) Request form: the *Max: Stream HBO, TV, & Movies* app redirected users to a request form where they needed to input their first name, last name, email, and country to delete the account. 2) Account deletion page with buttons: the *DramaBox - Stream Drama Shorts* app redirected users to an account deletion page, allowing them to click the “Delete Account” button to complete the process directly. 3) Step-by-step tutorial: the *X* app provided a step-by-step tutorial, guiding users through the account deletion process on the web. None of them mentioned how to delete accounts within the app.

3.4 Account Deletion Functionality Issues

Definition: An app provides an in-app path or web-based account deletion method. However, users could not successfully delete their accounts following their account deletion instructions. We will discuss how we detect these issues in Section 4.4.

Finding: There are three cases where users can not delete their accounts: 1) the app explicitly states that it can not delete the account due to certain reasons, 2) the account can not be deleted because of technical issues, and 3) the app claims the account has been deleted, but users can still log in using the same credentials or encounter a message stating that the account already exists when attempting to re-register. For example, the *Shein* app technically provided a web-based

account deletion method. After logging in and clicking the account deletion link, users were presented with a note and an “Apply to Delete Account” button, as shown in Figure 5a. However, upon clicking the button, a message appeared stating, “We are legally obligated to retain your data temporarily and cannot fulfill your request” as shown in Figure 5b, refusing to delete user accounts. We also found one app’s account deletion link (*ESPN*) directed users to the parent company’s account deletion website (*Disney*), but the account could not be found during the deletion process. While we successfully registered and logged in to *ESPN* using a *MyDisney* account on both the apps and web platforms, attempting to delete the account through the provided link resulted in an “invalid ID or password” error despite entering the correct credentials.



(a) The initial account deletion page. (b) Pop-up window after clicking the account deletion button.

Figure 5: Account deletion page of the *Shein* app. (a) displays the initial account deletion interface featuring a “Apply to delete account” button. (b) shows a pop-up window stating that the account cannot be deleted after clicking the button.

4 DELETETRACKER Design

Overview: Motivated by the preliminary findings from our pilot study, we develop a tool named DELETETRACKER to semi-automatically detect various account deletion issues as defined in Section 3. Figure 6 presents the design overview of DELETETRACKER, which comprises four modules: Account Deletion Link Crawler (Section 4.1), Account Deletion Link Analyzer (Section 4.2), Account Deletion Method Classifier (Section 4.3), and Account Deletion Validator (Section 4.4).

The Account Deletion Link Crawler (ADLC) automatically collects account deletion information from both the Google Play mobile app and website, preparing for subsequent analysis. Next, the Account Deletion Link Analyzer (ADLA) examines the availability, cross-platform inconsistency, accessibility, and similarity of account deletion links (Section 3.2). The Account Deletion Method Classifier (ADMC) detects content issues described in Section 3.3 by classifying account deletion methods, including missing account deletion method, alternative account deletion method, missing web-based account

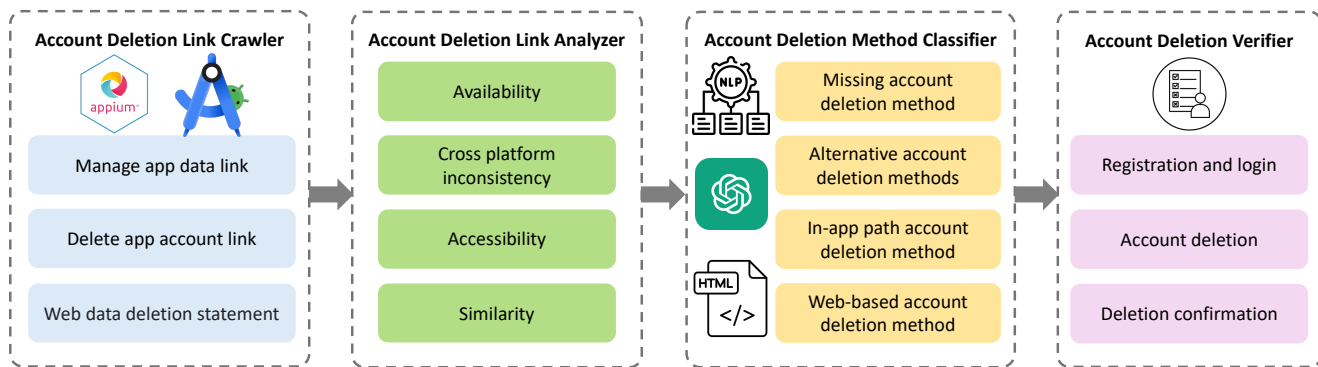


Figure 6: Overview of DELETETRACKER.

deletion method and missing in-app path account deletion method. Finally, the Account Deletion Verifier (ADV) verifies whether users can successfully delete their accounts using the methods provided in the apps’ account deletion links (functionality issue in Section 3.4). The Account Deletion Link Crawler and Analyzer are fully automated, the Account Deletion Method Classifier is semi-automated, and the Account Deletion Verifier is performed manually due to the significant difference between account deletion steps in apps.

4.1 Account Deletion Link Crawler

The Account Deletion Link Crawler (ADLC) automatically crawls “Manage app data” and “Delete app account” links from Google Play mobile apps (Section 4.1.1), and the “You can request that data be deleted” statement from Google Play websites (Section 4.1.2).

4.1.1 Mobile App Crawler

For the Google Play mobile app, ADLC retrieves the app ID (e.g., `com.amazon.avod.thirdpartyclient`) from Andro-Zoo [28, 30] and uses `google-play-scraper` [10] to collect additional app information, including the app name (e.g., Amazon Prime Video) and genre. We need both app ID and name to accurately identify an app on the Google Play because using only app ID may lead to sponsored apps, and using only app name can return apps with the same name. After obtaining the app names, we use Appium³, which provides UI automation, along with a virtual machine (Google Pixel 9) on Android Studio, to simulate human interactions with mobile devices. During interactions, we need to click and scroll through various elements to navigate to the correct page. To locate each required element, we identify its XPath which serves as the “coordinate” of the element in the app’s UI. We utilize the Appium Inspector⁴, a tool for visually inspecting and interacting with mobile apps to retrieve the

corresponding XPath. For example, when we click the search button of Google Play in the virtual machine, Appium Inspector displays its XPath (i.e., `android.widget.TextView[@text="Search"]`).

ADLC starts by ensuring the Google Play homepage is open by setting `com.google.android.finsky.activities.MainActivity` as the entry point, preparing for a new app search. Then, it clicks on the search button and inputs the app ID. ADLC then matches the app name and clicks on it to navigate to the corresponding app page. After the app page loads, it scrolls down to locate the “Data Safety” section and clicks to enter it. Within the DSS, the crawler scrolls further to find the “Go to developer’s site” which is the clickable element that contains the data deletion link. The crawler can find and click on it to extract the URLs. Finally, our system copies these URLs and saves them into a text file, and then returns to the homepage to prepare for the next app search. ADLC repeats this process for every app and saves the ‘Delete app account’ and ‘Manage app data’ links unless an app doesn’t provide them.

However, an app may contain 0, 1 or 2 “Go to developer’s site” links, corresponding to the “Delete app account” and “Manage app data” sections. Without analyzing the context, it is hard to match the link with its corresponding section. To resolve this issue, we first detect the count of “Go to developer’s site” links searching for `android.widget.TextView` elements with the text “Go to developer’s site.” If the count is one, we check the existence of the XPaths for “Delete app account” and “Manage app data.” If either of these sections exists, the link is associated with the corresponding section. If there are two “Go to developer’s site” links, their XPaths are indexed (e.g., `android.widget.TextView[@text="Go to developers site"][1]`). The links are bound to their respective sections in a predetermined order: the first link ([1]) is paired with “Delete app account” and the second link ([2]) is paired with “Manage app data”. This method ensures accurate pairing of links to their respective sections.

³<https://github.com/appium/appium>

⁴<https://github.com/appium/appium-inspector>

4.1.2 Web Crawler

For the Google Play website page, we use `requests` [18] and `BeautifulSoup` [4] to implement a web crawler to check for the presence of data deletion statement “*You can request that data be deleted*”. We automatically open the webpage for each app (https://play.google.com/store/apps/details?id=app_id) and search for the sentence.

4.2 Account Deletion Link Analyzer

The Account Deletion Link Analyzer module aims to analyze the account deletion links (ADL) from the following four aspects:

- How many apps provide the ADL? - Availability
- Is there any inconsistency in data deletion between the Google Play mobile app and the website? - Cross-platform inconsistency
- How many ADLs are inaccessible? - Accessibility
- How many apps share the same or similar ADLs? - Similarity

Availability: For each app, if the ADLC does not detect the “Go to developer’s site” link under the “Delete app account” section, we consider it a potential missing ADL.

Cross-platform inconsistency: For each app, ADLA compares the results of ADLC’s mobile app crawler and web crawler. If the web crawler detects the statement “*You can request that data be deleted*”, but the mobile app crawler does not detect that an app has either “delete app account” or “manage app data” links, we consider this a cross-platform inconsistency, and vice versa.

Accessibility: After obtaining ADLs, we first verify whether a URL starts with `http` or `https`. If it does not, we prepend the `https` and `http` prefixes to the URL. Then we use the `requests` library to send a request to the URL and get the response. If the response status code is 200, we consider the link as accessible.

Similarity: We first identify whether any apps share identical URLs for account deletion. Next, we analyze each part of the URLs to determine if they belong to the same parent domain. We split the URLs by the `/` delimiter and then compare the first part of each URL to determine whether they share the same domain. Typically, the first part of a URL after the prefix (e.g., `http://`) represents the company domain name. If the URLs are identical, they may potentially point to shared resources, such as a privacy policy or help center, which might lack specific account deletion information. If the URLs are similar, there could exist potential functional issues. For example, as discussed in Section 3.4, *ESPN*’s account deletion link redirects to Disney’s account deletion page. While we

successfully registered and logged in to *ESPN* using a *MyDisney* account, we were unable to delete the account through the provided link.

4.3 Account Deletion Method Classifier

The Account Deletion Method Classifier (ADMC) module aims to classify the methods each app provides for account deletion. Our analysis covers 4 categories: missing account deletion method, alternative account deletion methods, in-app path, and web-based account deletion.

For each account deletion link, we first use the `requests` library to send an HTTP request to fetch web content and use `BeautifulSoup` to parse the HTML. Next, we apply the tool `HtmlToPlaintext` [14] to convert HTML to plain text, making the content easier to analyze in the following steps. For each app, we apply two methods to classify account deletion methods: our developed tool and a prompted large language model (LLM). Due to the diversity of natural language, we adopt a semi-automatic approach for classification, which involves using automated tools while incorporating human intervention. We will have human intervention only in the following three scenarios: 1) the link requires login or registration; 2) the conclusions from the two methods are inconsistent; and 3) the content is not in English.

4.3.1 Missing Account Deletion Method

First, two annotators establish the initial word sets related to account deletion based on our pilot study (32 apps provide account deletion methods). We extract account deletion sentences and derive initial sets of relevant words including a verb set [“delete”, “close”, “cancel”] and a noun set [“account”, “profile”]. These sets are further expanded through human intervention when the results from NLP and LLM are different. As a result, we expand the word sets to include an additional verb set [“remove”, “erase”, “withdraw”, “end”] and noun set [“user”, “gameid”, “userid”]. In accordance with Google Play’s account deletion requirements, we do not consider “temporary account deletion”; therefore, keywords like “deactivate”, “freeze” and “suspend” are excluded from our word sets. To ensure a comprehensive detection, we account for different grammatical forms of words in both sets (e.g., “delete” → “deletes”, “deleted”, “deletion”, “deleting”) by applying lemmatization to normalize the text to its base forms. A link is considered to include account deletion information if at least one sentence contains words from both the verb set and the noun set. Otherwise, it is categorized as the missing account deletion method. We apply the `spaCy` [19], which is a Natural Language Processing (NLP) library, to tokenize sentences, apply lemmatization, identify matching keywords from the predefined word sets, and flag sentences that contain both verb and noun matches. Note that documents like pri-

vacy policies may also provide account deletion information. Therefore, the presence of account deletion information does not necessarily mean that the link includes a functional account deletion method. For example, *Brawl Stars*'s ⁵ account deletion link states, “It is possible to request the deletion of all the information we have collected about your game account.” This sentence contains account deletion information. However, beyond this statement, the link does not provide any specific account deletion method or detailed instructions for users to complete the process.

Since the presence of account deletion information does not guarantee the account deletion methods, we further classify account deletion methods in the following sections. If an app does not fit into any of the defined categories (*i.e.*, missing in-app path, missing web-based or alternative account deletion method as described in Section 3.3), we still consider such links as missing account deletion methods.

4.3.2 In-app Path Account Deletion Method

Similar to Section 4.3.1, to identify the in-app path account deletion method, two annotators first manually compile a list of commonly used phrase patterns from 12 apps that provide such a method in our pilot study. These phrase patterns are further expanded through human intervention. Normally, an in-app path account deletion is often presented as a step-by-step tutorial, guiding users through the process of deleting their accounts. This may include instructions such as “open the app” or “tap account settings,” as shown in Listing 1. For apps that are not flagged as missing account deletion methods, we use a dependency parsing method to identify the in-app path account deletion. Dependency parsing is an NLP technique that analyzes grammatical relationships between words in a sentence. This method enables us to extract meaningful action-object pairs (*e.g.*, “open app,” “tap settings”) that indicate potential account deletion paths within the app.

```
1 phrase_patterns = [  
2     r"open\s+\w+\s+game", # "open the game"  
3     r"launch\s+\w+\s+app", # "launch MacDonald app"  
4     r"tap\s+\w+\s+profile", # "tap user profile"  
5     r"tap\s+\w+\s+settings" # "tap account settings"  
6 ]
```

Listing 1: Phrase patterns example. \s+ matches one or more whitespace characters, such as spaces, tabs, or line breaks. \w+ matches one or more word characters, including letters, digits, or underscores.

Token and lemma analysis: Each sentence is tokenized into individual words, and their base forms (lemmas) are extracted to account for variations such as “opening” and “opened” for the verb “open.”

Dependency relations: We focus on specific dependency labels, such as: (1) *doobj* (direct object): Identifies objects

⁵<https://support.supercell.com/brawl-stars/en/articles/gdpr-request-deletion-of-your-personal-data.html>

directly acted upon by verbs (*e.g.*, “open app”). (2) *prep*, *pobj*, *advmod* (prepositional object and modifiers): Identifies additional context, such as “in-app” or “via the app.”

Pattern matching: Dependency relationships are analyzed to match predefined patterns, such as *Verb + Direct Object* → “open Netflix app” and *Verb + Prepositional Phrase* → “delete account via app”.

For each app, if the in-app path-related phrase patterns are detected and contain account deletion information, we classify it as having an in-app path account deletion method.

4.3.3 Web-based Account Deletion Method

The web-based account deletion method can be further divided into three categories: step-by-step tutorials, request forms, and direct navigation to an account deletion page with clickable buttons.

Step-by-step tutorials: To detect step-by-step tutorials, we apply a similar method to that used for detecting in-app path methods as described in Section 4.3.2.

Request form: We identify request forms on web pages by analyzing the HTML structure of given URLs. It searches for <form> elements and extracts input fields, including <input>, <textarea>, and <select>. These fields typically include email, user ID, account ID/name, password, and the reason for account deletion.

Account deletion page with clickable buttons: The method involves extracting all <button> elements and clickable links (*i.e.*, <a> tags with href attributes) from the page. Then a keyword matching is performed to check if the text within these buttons or links contains relevant terms such as “delete”, “withdraw”, “erase”, “terminate”, which may indicate a potential account deletion option.

4.3.4 LLMs Classification

To improve the accuracy of the Account Deletion Method Classifier and detect alternative account deletion methods, we leverage LLMs as a complementary step after our rule-based methods. Specifically, we utilize the OpenAI API and design prompts to guide GPT-4 in identifying whether the content corresponds to the missing account deletion method, alternative account deletion method, in-app path account, and web-based account deletion. In the prompt, we first provide definitions of 4 account deletion methods. Subsequently, we ask the LLMs to perform classification based on a three-step instruction. The detailed prompts are illustrated in Figure 7. After obtaining answers from the LLM, we compare the results with those generated by our developed tools. In cases where discrepancies arise, two annotators manually perform the classification. The classification criteria is established based on our pilot study, with reference to the textual definitions and examples in Section 3.3. Any disagreements

Prompting Design

Prompt: You are tasked with reviewing the content and classifying the method(s) it describes for account deletion method (ADM). There are four possible categories:

1. Missing ADM: The content does not include any content related to ADM.
2. In-app path ADM: The content describes the specific steps or navigation paths users can follow within an app (on Android, iOS, or mobile platforms) to delete their account.
3. Web-based ADM: The content provides a direct URL or interface where users can input their information and delete their account online.
4. Alternative ADM: The content provides alternative methods such as email, telephone and so on.

Tasks:

1. Review the following content.
2. Determine which of the above categories apply (it can be one or more).
3. Output the classification as follows: [Missing], [Alternative-xxx], [In-app path], [Web-based] or a combination if multiple methods are mentioned.

Content: You can either use the in-game method mentioned below to delete your account, or you can send a mail to privacy@fugo.com.tr with your credentials to request deletion of your account. In the main menu of the game, you will see the 'Settings' button on top right. In the Settings page, you can use the 'Delete Account' button on the bottom to delete your account in the game.

Answer: [In-App Path, Alternative-email]

Figure 7: Prompting design. The content is from the account deletion link of the *Words of Wonders: Crossword* app.

between the two annotators are resolved through discussion and more details will be provided in Section 5.5.1.

4.4 Account Deletion Verifier

To verify whether each app truly deletes user accounts, we follow the account deletion steps provided on their URLs. Due to the significant differences between apps, it is difficult to implement a tool-based automatic approach. Thus, this module is done manually. Two annotators perform the tasks of registration, login, account deletion, and confirmation of deletion to mitigate potential human errors or biases. If they get different results, they repeat the process together, check each action carefully, and make a final decision after discussion.

For consistency, each annotator uses different fixed virtual user profiles for account registration and login, including email address, password, full name, and other attributes. More details of virtual users are provided in Table 4 in Appendix B. We prioritize using the email-and-password registration method, as it facilitates more reliable verification of account deletion. However, for apps that only support third-party login methods (e.g., Google sign-in), we proceed with the alternative verification option. Some apps provide unique account identifiers, such as user IDs or account numbers, thus we can use them to verify account deletion. For apps that lack such unique identifiers, we create manual usage traces (e.g., adding bookmarks in browsers or liking songs in music players). This allows us to verify whether these traces still existed in the account upon re-login and determine if the account was truly deleted.

Since certain apps require 7, 14, or 30 days to complete the deletion process, we verify these cases one month after initiating deletion. We refer to the criteria that previous work [46] uses to determine whether account deletion is successful, including: 1) **For accounts registered via email and password:**

we first attempt to re-login using the same credentials. If the re-login is successful, we conclude that the account is not deleted. If the login fails, we attempt to re-register using the same credentials and user information. If the system indicates that the email or username is already registered, or that the account belongs to a previously deleted user, we conclude that the account has not been successfully deleted. 2) **For accounts logged in via Google sign-in:** we attempt to re-login using the same Google account. If the account contains previously created usage traces, we consider the account deletion unsuccessful.

5 Results and Findings

We implemented DELETETRACKER in Python. We applied DELETETRACKER to check Google Play apps. The overall results of DELETETRACKER are shown in Table 1.

Module	Issue/compliance	# of apps
Account Deletion Link Analyzer (Automated)	Missing account deletion link	291
	Cross-platform inconsistency	65
	Inaccessible link	25
	Same/similar link	197
Account Deletion Method Classifier (Automated + Manual)	Missing account deletion method	98
	Alternative account deletion method	35
	Missing web-based account deletion	68
	Missing in-app path account deletion	251
Compliance without issue		42
Account Deletion Verifier (Manual)	Account deletion failure	12

Table 1: Results of DELETETRACKER.

5.1 Account Deletion Link Retrieval

Due to the massive number of apps published on the Google Play platform and the fact that our tool is not fully automated, we only selected 1,000 apps from the AndroZoo [28, 30] dataset, which is open source and contains 24,855,480 APKs. We chose the top 1,000 apps based on their rating count because other statistics, such as download counts, only provide rough estimates (e.g., 100M or 10M downloads). We collected apps' metadata via API provided by AndroZoo and used the `google-play-scraper` library to retrieve the app names in October 2024. With the help of Account Deletion Link Crawler, we successfully retrieved information for 863 apps out of 1,000 APK names. The remaining 137 apps could not be retrieved because 65 apps were unavailable via `google-play-scraper`, 70 were region-restricted, and 2 were unsupported due to virtual machine limitations.

Performance: For the 137 apps, we manually verified them and confirmed that even with manual methods, we were unable to obtain their account deletion links. Therefore, we

excluded them from our 1,000 apps dataset. Among the 863 apps, 572 apps have an account deletion link. We manually opened their account deletion links on an Android Studio Virtual Machine, copied the URLs into a document, and compared the copied URLs with those URLs crawled by DELETE-TRACKER to verify whether they are the same. The Account Deletion Link Crawler achieved 100% precision. We also randomly sampled 100 apps from the 863 apps to estimate the recall of the Account Deletion Link Crawler. We used these 100 apps for recall calculation in the modular evaluation of DELETE-TRACKER and denote them as *ME*₁₀₀. More details about the modular evaluation will be discussed in Section 5.5.2. Our crawler achieved 98% recall and it failed to retrieve the account deletion links for 2 apps due to their uncommon page layout.

5.2 Account Deletion Link Analysis

Missing account deletion link: Among the 863 apps, we found that 572 apps provided an account deletion link and 291 apps didn't. Among the 291 apps missing an account deletion link, only 26 apps provided a “Manage App Data” link with a clear statement: “There’s no account to delete for this app. You can submit a request to the developer to delete the data collected through this app and linked to you.” The remaining 265 apps had the potential to violate legal regulations and Google Play’s app account deletion requirements. In this work, we did not delve into the reasons why these 291 apps fail to provide an account deletion link. Both GDPR and Google Play’s app account deletion requirements allow for exemptions in certain cases, and determining whether these conditions qualify for such exemptions could be a legal matter.

Longitudinal analysis: Google Play claims that by December 7, 2023, developers will no longer be able to publish a new app or update the app if account deletion information is not provided, and non-compliant apps may face enforcement actions, *e.g.*, removal from Google Play, after May 31, 2024. We conducted our first analysis in October 2024, half a year after Google Play’s deadline for developers to comply with these requirements. To evaluate how many apps have made the necessary changes since our initial data collection and analysis, we re-collected account deletion links for the 291 apps missing an account deletion link after one month. We found that only one app (*i.e.*, *Bitmoji*) added an account deletion link to its DSS. We speculate that the addition of the account deletion link was in response to Google Play’s enforcement actions.

Cross-platform inconsistency: Among the 291 apps missing account deletion links, we identified 65 apps with inconsistencies between the Google Play website and mobile app: these apps state that users can request data deletion on the Google Play website, but on the Google Play mobile app, they do not provide either “Delete app account” or “Manage app data” links. In contrast, all the 572 apps that provided data deletion

links in the mobile app also stated that users could request data deletion on the webpage without inconsistency.

Inaccessible links: Among the 572 apps providing an account deletion link, we found that 25 links were invalid and could not be accessed. For example, the *WEBTOON* app provided an inaccessible account deletion link⁶.

Same/similar links: Among 547 apps with accessible account deletion links, we found that 130 apps are sharing 38 account deletion links with other apps. This situation is common among large companies, such as Google (*e.g.*, Gmail, Google Maps, YouTube), Facebook (*e.g.*, Facebook, Messenger, Meta Business Suite), and Amazon (*e.g.*, Amazon Kindle, Amazon Prime Video). The largest cluster is Google-related apps, where 21 apps share the same account deletion link (<https://support.google.com/accounts/answer/32046>). From the content of this link, we observed that, except for Gmail, there are no app-specific accounts, and users can only delete their entire Google account if they want to delete the account for one app. In addition to large companies like Google which develop multiple apps using a unified account system, we also found that gaming companies frequently share a single account deletion link. Unlike Google, 37 gaming apps include a dropdown menu on their deletion page, allowing users to select a specific game account to delete as shown in Figure 8. Surprisingly, this webpage supports the deletion of accounts for 638 games.

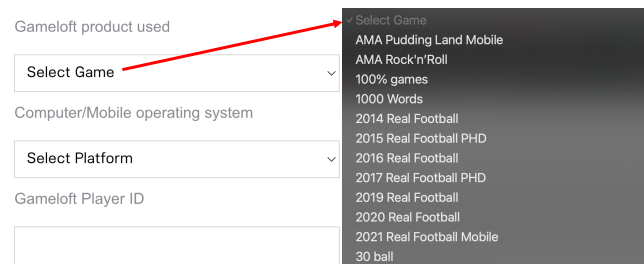


Figure 8: Gameloft games’ account deletion page.

We also found that apps from the same company often have similar but not identical account deletion links (*i.e.*, same company domain). Specifically, we identified 67 apps sharing 21 domains belonging to the same company. Within these apps, we found an interesting phenomenon that some links provide valid account deletion information, while others do not. For example, in the case of *Opera Mini: Fast Web Browser* and *Opera Browser with AI*, their account deletion links are similar but lead to different outcomes: the account deletion link for *Opera Mini: Fast Web Browser* (auth.opera.com/account/authenticate/email) does not provide effective account deletion information. In contrast, the link for *Opera Browser with AI* (auth.opera.com/account/delete-profile) offers a clear and functional account deletion method.

⁶<https://www.webtoons.com/en/account/delete>

This demonstrates that even for products from the same company, the implementation of account deletion links can vary significantly, leading to inconsistencies in user experience.

Performance: After manually verifying all the detected issues, the Account Deletion Link Analyzer achieved 96.3% precision. We used *ME*₁₀₀ to check whether they contained any issues and whether those issues could be detected by DELETETRACKER. More details about how we annotated apps will be discussed in Section 5.5.1. Finally, the Account Deletion Link Analyzer achieved 98.1% recall. Most false positives and false negatives occurred in the accessibility check, primarily due to occasional network instability and variable response times. For example, the account deletion link for *OZON* required redirection and took additional time to fully load. Although we set a 10-second timeout threshold, the page failed to respond within that window, resulting in the link being incorrectly marked as inaccessible.

5.3 Account Deletion Method Classification

Using the Account Deletion Method Classifier, we automatically classified the account deletion methods for 323 apps among 547 apps. For apps that cannot be automatically processed, we manually classified them. In total, we successfully classified the account deletion methods for 494 apps from 547 apps. The remaining apps could not be classified due to 18 apps requiring sensitive personal information to register and access the account deletion webpage, 26 only supporting in-app registration, 2 being non-English, and 7 having non-functional registration processes.

Missing account deletion methods: Among the 494 apps, 19.8% (98) apps provided a link that doesn't contain any information related to account deletion. These links mainly have the following issues: 16 links directed users to FAQs or Help Centers containing extensive content, making it difficult to locate account deletion instructions, or requiring multiple steps to access correct account deletion page; 25 links led to privacy policies or terms of use, which often mention data rights, including the right to delete accounts, but did not provide specific instructions of account deletion; 39 links redirected to the account profile page or app homepage without clear steps, requiring users to navigate specific sections to delete their accounts; the other 18 links indicated that these apps did not provide account deletion functionality or only supported temporary measures, such as account deactivation or freezing.

Providing alternative account deletion method: We found that 10.5% (52/494) of apps required users to send an email to their developer team for account deletion, and 2.4% (12/494) of apps asked users to make a phone call or visit a physical branch, a practice commonly seen in banking and financial apps. Figure 10 in Appendix A shows an example of *Jaz-zCash* which only provided three alternative account deletion methods. Although these apps provided a method for users to delete their accounts, they still didn't satisfy Google Play's

app account deletion requirements. Therefore, we considered them as missing the required account deletion methods.

Missing web-based account deletion: We found 13.8% (68/494) apps that only provided an in-app path account deletion method and didn't provide the web-based account deletion method, which didn't comply with Google Play's app account requirements.

Missing in-app path account deletion: There were 50.8% (251/494) apps that only provided a web-based account deletion method, which also violated Google Play's app account deletion requirements. Of the 251 apps, 55 apps offered step-by-step tutorials guiding users to find the account deletion method, 127 apps required users to fill out a form, and 70 apps directed users to an account deletion page where the process can be completed by clicking buttons.

No compliance issues detected: DELETETRACKER found that 8.5% (42/494) of the apps provided both in-app and web-based account deletion methods, which fully complied with Google Play's app account deletion requirements.

Performance: Account Deletion Method Classifier achieved 84.5% precision and 83.0% recall. To evaluate the stability of the GPT-4, we conducted a second-round experiment and its repeatability is 98.3%. There are several reasons for misclassification. First, our tool failed to extract content from webpages using BeautifulSoup or HtmlToPlaintext, resulting in no issues being detected. Second, some webpages required user interactions to access the full account deletion instructions, such as clicking buttons to reach the account deletion page or expanding collapsible sections, which cannot be handled by our tool. Additionally, ChatGPT also misinterpreted the intent or context of the webpage content in a few cases.

5.4 Account Deletion Verification

Due to the significant differences between apps, it is difficult to implement a tool-based automatic approach. Therefore, for the 363 apps providing either in-app path or web-based account deletion method, we manually registered, deleted, relogged in, and re-registered accounts to verify whether each app truly deletes user accounts, as they claim in the DSS. Since some apps could not be registered due to requirements such as a phone number, credit card information and subscription or region restrictions, we successfully registered accounts for 182 apps. During the deletion process, we encountered issues such as being unable to find the deletion page, needing to send an email, or being informed that no user data had been collected. Finally, we successfully submitted 176 account deletion requests. Because this module relies entirely on manual verification, traditional evaluation metrics (*i.e.*, precision and recall) are not applicable. Instead, the two annotators cross-validated each other's work to ensure accuracy and consistency.

Finally, we found 12 apps failed to delete the account as promised in the account deletion links. 2 apps have over 500M

Issue	App Name
Functionality issues	Glovo: Food & Grocery Delivery, Nebulous.io
Re-login issues	Six Pack in 30 Days, Step Counter - Pedometer, Lose Weight App for Men, Phoenix - Fast & Safe, King James Bible - Verse+Audio
Re-register issues	Yandex Maps and Navigator, Yandex Navigator, MyFitnessPal: Calorie Counter, Geometry Dash Lite, Reddit

Table 2: Results of Account Deletion Verification.

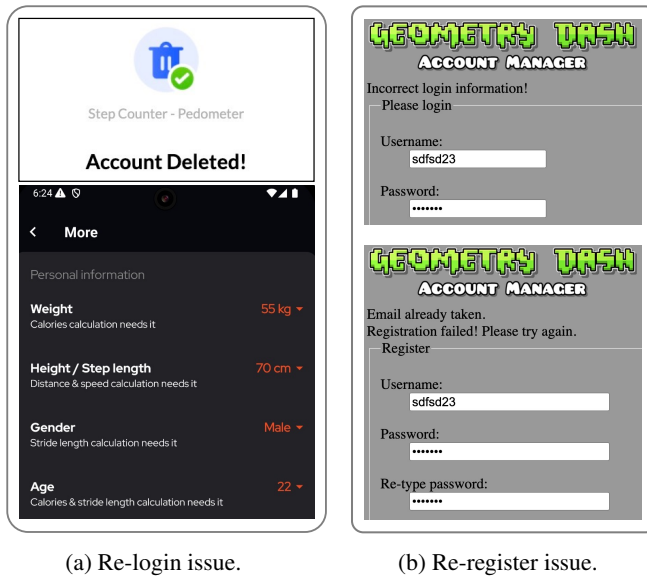


Figure 9: Examples of unsuccessful account deletions.

downloads, 7 have over 100M downloads, and the other 3 have more than 50M downloads, showing their popularity. Table 2 lists the names of these apps and their issues. First, we identified two apps with functionality issues. For example, in the *Nebulous.io*, we were unable to submit the account deletion request because the “Delete” button showed an error message after clicking. Another issue is related to re-login (5 apps). For instance, in the *Step Counter - Pedometer* app, we manually left usage traces by setting the weight and age. After submitting an account deletion request and signing in again, our previously set information remained in the account, as shown in Figure 9a. This indicates that the account and associated data had not been properly deleted. For re-register issues (5 apps), the *Geometry Dash Lite* app stated that the account would be deleted 27 days after the deletion request was raised. However, when we attempted to re-register with the same email address after the stated data retention period, it displayed “Email already taken” (Figure 9b). This indicated that the account information (at least email) still exists within their system despite their claim of deletion.

For the apps that claim to delete user accounts but don’t actually do that, if they are published on the European market-

place, they not only violate platform requirements but also do not comply with legal regulations, such as GDPR. Article 17 of GDPR states that “The data subject shall have the right to obtain from the controller the erasure of personal data concerning him or her without undue delay, and the controller shall have the obligation to erase personal data without undue delay”. Therefore, the apps refusing to delete data or pretending to delete user accounts but still keeping data in the database could be a violation of GDPR, which may lead to fines from the government, similar to how Google was fined [8]. Meanwhile, these apps that have already been published in the US marketplace could violate regulations in the U.S., such as the California Consumer Privacy Act (CCPA) [5], the Virginia Consumer Data Protection Act (VCDPA) [25], the Colorado Privacy Act (CPA) [6], the Connecticut Data Privacy Act (CT-DPA) [20], and the Utah Consumer Privacy Act (UCPA) [24].

5.5 Evaluation

5.5.1 Ground Truth Annotation Process

To establish the ground truth and minimize bias, two annotators independently conducted the annotation. The annotation was conducted using two Android Studio virtual machines (emulating Pixel 9 devices). Following a procedure similar to that used in the pilot study, each annotator acted as a normal user to examine the DSS for each app. The annotators followed the definitions of compliance and non-compliance issues as described in Section 3. For each app, both annotators were required to select one option for the following parts:

Annotation Criteria (per App)

- 1. Missing account deletion link:**
 - ☐ [url] ☐ missing
- 2. Inaccessible account deletion link:**
 - ☐ accessible ☐ inaccessible
- 3. Cross-platform inconsistency:**
 - ☐ consistency ☐ inconsistency
- 4. Non-compliant account deletion method:**
 - ☐ missing account deletion method
 - ☐ missing in-app path method
 - ☐ missing web-based method
 - ☐ alternative account deletion method ☐ compliance

After independently annotating the data, the two annotators compared their results. For any inconsistencies, they held discussions to reach an agreement. Finally, for the inter-rater agreement score, we obtained Cohen’s Kappa values of 1 and 0.98 (almost perfect agreement) for the first two modules (Account Deletion Link Crawler and Account Deletion Link Analyzer), respectively. This is because they are based on objective criteria, *i.e.*, whether a link is available or accessible. In contrast, the Account Deletion Method Classification module requires subjective interpretation of account deletion link content, leading to a slightly lower agreement value (0.76:

substantial). For example, the app *Guns of Glory: Lost Island* says “Please contact our in-game customer service by clicking Setting and then Contact Help”, which is somewhat ambiguous and leads to disagreement between two annotators. One annotator interpreted it as an in-app path to reach customer service within the game interface, while the other annotator believed the final destination was an external customer service webpage, thus interpreting it as a web-based method. After further reviewing Google Play’s official definitions, annotators agreed that the instruction describes an in-app path for account deletion.

5.5.2 Performance

We performed both modular evaluation and end-to-end evaluation for DELETETRACKER, and the results are summarized in Table 3. More details about the number of apps used for evaluation are provided in Table 5 in Appendix C.1.

Type	Module Name	Precision	Recall
Modular Evaluation	Account Deletion Link Crawler	100%	98%
	Account Deletion Link Analyzer	96.3%	98.1%
	Account Deletion Method Classifier	84.5%	83.0%
End-to-end Evaluation	Issue-level end-to-end performance	89.5%	91.7%
	App-level end-to-end performance	88.8%	91.9%

Table 3: Performance of DELETETRACKER.

For the modular evaluation, two annotators annotated all the detected results to calculate the precision for each module, and used the randomly sampled 100 apps from the 863 apps (*i.e.*, $\mathbf{ME}_{100}$) to check whether any issues were missed to calculate the recall. More details on the breakdown of issues identified in $\mathbf{ME}_{100}$ are shown in Table 6 in the Appendix. We discuss the limitation and potential bias of our app sampling for recall calculation in Section 6.2.

For the end-to-end evaluation, we randomly sampled another 100 new apps (denoted as $\mathbf{EE}_{100}$, which is different from $\mathbf{ME}_{100}$) from all Google Play apps, *excluding* the 863 apps which were used in the development of DELETETRACKER, and manually labeled all account deletion issues they contained. We conducted the end-to-end evaluations at two different levels of granularity: the issue-level and the app-level.

Issue-level evaluation: In the issue-level end-to-end evaluation, we cannot directly multiply the performance metrics of each module to get the end-to-end performance because some apps may not pass through all the modules. For example, if an app does not provide an account deletion link, it is unnecessary to analyze whether it misses an account deletion method. Since one app may be associated with multiple issues, which is a multi-label classification task, we applied the micro-averaged precision and recall - a standard metric for this type of task [15]. Micro-averaging aggregates the contributions of all classes (or issues) to compute the overall performance, as shown in the following equations (i denotes

the index of an issue defined in Section 3):

$$\text{Micro-Precision} = \frac{\sum_i TP_i}{\sum_i (TP_i + FP_i)} \quad (1)$$

$$\text{Micro-Recall} = \frac{\sum_i TP_i}{\sum_i (TP_i + FN_i)} \quad (2)$$

Based on Equations (1) and (2), we computed the confusion matrix for each issue and summed them up to compute the overall precision and recall as shown in Table 8 in the Appendix. Since the issue-level evaluation is a weighted average of the performance of each component in a pipeline, if other modules have significantly higher precision and recall and larger numbers of instances, they will dominate the final score. As a result, the overall micro-average score can be higher than the score of a single module. More details about the definition of confusion metrics for the issue-level end-to-end evaluation are provided in Appendix C.2.

App-level evaluation: We assigned a single TP, FP, FN, or TN label to each app, based on whether all of its issues were correctly identified. In total, 83 apps (*i.e.*, TP + TN) were correctly identified, while 17 apps were incorrectly handled. These failures include cases where the account deletion link was accessible but DELETETRACKER mistakenly labeled it as inaccessible, or instances where DELETETRACKER failed to retrieve the account deletion link or misclassified the account deletion method. We define the confusion metrics as follows:

- TP (79 apps): DELETETRACKER detects all issues correctly, and the detected issues match the ground truth.
- FP (10 apps): DELETETRACKER detects more issues than those present in the ground truth, or it misclassifies the type of an issue.
- FN (7 apps): DELETETRACKER misses one or more issues compared to the ground truth.
- TN (4 apps): The app does not contain any issue, and DELETETRACKER does not detect any issue.

Some apps, such as those with missing account deletion links, do not reach the Account Deletion Method Classifier (ADMC). Since such apps account for a relatively large portion of TPs, the total number of TPs is higher, resulting in a higher app-level end-to-end performance compared to that of the individual ADCM module.

Both end-to-end evaluations were conducted directly on the set of 100 apps $\mathbf{EE}_{100}$. As shown in Table 3, DELETETRACKER achieved a precision of 89.5% and a recall of 91.7% in issue-level end-to-end evaluation, and 88.8% precision and 91.9% recall in app-level end-to-end evaluation.

6 Discussion

6.1 Implications

Our developed tool DELETETRACKER and findings from apps have implications for various stakeholders, including app plat-

forms, developers, legal regulators, researchers and users.

First, the Google Play platform claims that non-compliant apps could be removed from the Google Play platform after May 31, 2024. However, our longitudinal analysis shows that this enforcement has not occurred (Section 5.2), possibly due to the lack of a tool to perform large-scale analysis. Our findings and identified issues were acknowledged by Google which assigned the issue a Medium (S2) severity level, indicating that Google recognized it as an unsolved issue. Therefore, DELETETRACKER can potentially help Google identify account deletion issues before apps are released, or perform large-scale policy violation detection in apps already available on the Google Play platform, similar to our analysis (we only analyzed 1,000 apps and 8.5% of apps complied with Google Play’s account deletion policy). Likewise, other app platforms, such as Apple’s App Store, could deploy similar tools to detect account deletion issues on their platforms.

Second, for app developers and companies, our findings may encourage them to prioritize easy and transparent account deletion processes to comply with Google Play’s app account deletion requirements. We have notified developers about the issues identified in their apps and are currently waiting for their responses. One possible reason for non-compliant apps is that some developers may be unaware of the policy requirements, as noted in previous research [39, 44]. Other developers may misunderstand the policies, such as assuming the account deletion link is not required if their app does not collect personal data [12, 13]. By having a clearer understanding of platform policies and making improvements, developers can build user trust and reduce potential violations.

Third, legal stakeholders, such as regulators of GDPR and CCPA, could also adopt our tool to identify if an app violates legal regulations and fails to erase users’ personal data (*e.g.*, violation of GDPR article 17, as discussed in Section 5.4). Fourth, we have shared our tool and findings (<https://zenodo.org/records/15610882>) with the community, which could facilitate future research in this area. Researchers could discover additional issues related to the account deletion procedures or develop more advanced automated tools to detect such issues in apps at scale. Finally, our tool can assist users who are highly concerned about privacy control to quickly identify whether the apps they use provide account deletion functionality. Also, our findings may encourage users who previously requested account deletion to verify whether their accounts have been properly deleted. Through insights presented in this work, users become more aware of the data deletion options and user privacy rights. Then they can better protect their privacy and personal information when using apps in the future.

6.2 Limitations

With DELETETRACKER, we uncovered various issues related to account deletion in Google Play apps. However, there are sev-

eral limitations in our work, which can be further improved.

First, the dataset we used for our analysis and evaluation is relatively small. We conducted our analysis on only 1,000 popular apps using DELETETRACKER. While our study on these apps revealed many common account deletion issues, expanding the dataset to include a more diverse selection of apps may help uncover more issues. In addition, our evaluation is not fully comprehensive as we sampled only 100 apps from the 863 apps, instead of from all the app population, to compute recall for the modular evaluation. We followed this approach to ensure that annotators could access the app and evaluate issues consistently and fairly. However, this may introduce potential sampling bias and the reported recall values may not accurately reflect DELETETRACKER’s performance across the entire app population.

Second, the efficiency of our crawler tool could be further improved. Unlike web crawlers, our mobile crawlers need to simulate human interactions to extract account deletion links, resulting in a slower process. Also, our crawler failed to retrieve some apps due to region restrictions, unavailability via Google search, and lack of virtual machine support.

Third, due to the varying layouts, account registration methods, and account deletion methods across different apps, it is hard to develop a fully automated tool to complete the entire process described in Section 4.4. While we mainly focus on disclosing the existence of various issues about account deletion links in this paper, the optimization and automation of DELETETRACKER could be implemented in future iterations. We will explore alternative methods, such as analyzing APK files, to efficiently extract relevant information in the future.

Fourth, we conducted our account deletion verification on all apps after 30 days, as apps may implement different data retention periods after account deletion. However, we did not consider short retention windows (*e.g.*, 3 or 7 days) when we verified account deletion, which may lead to some cases being missed. We plan to verify the apps based on their corresponding data retention period.

Finally, we only focused on Google Play and didn’t consider other platforms like the Apple App Store. Given the similar account deletion requirements on these platforms, we plan to apply a similar method to detect account deletion issues within the Apple App Store.

7 Related Work

Privacy compliance of apps: Privacy policy serves as an important way for companies and developers to provide privacy disclosures to users. However, several studies have identified privacy policy issues in different platforms [27, 40, 48, 49]. To enhance the readability of privacy notifications and ensure privacy compliance, Kelley proposed the “privacy nutrition label” [37], which has gained popularity in recent years with more platforms adopting it (*e.g.*, Apple Privacy Label, Google Play DSS). However, the quality of apps’ privacy labels varies.

Rodriguez *et al.* [45] found that 66.5% of the same apps have discrepancies in their privacy labels between Apple’s App Store and Google Play. Furthermore, discrepancies were also identified between the privacy labels and privacy policies of the same app [29]. Researchers also identified inconsistencies between actual app code or behaviors and privacy labels [31, 33, 38]. Longitudinal studies further highlight the issues, such as low adoption rates of privacy labels, minimal updates, and incomplete disclosures [32, 33]. However, none of them have analyzed account deletion issues in Google Play’s DSS, and our work fills this gap.

Account deletion: Liu *et al.* [42] found that users often keep “zombie” app accounts due to a lack of awareness and face significant challenges during the deletion process. Schaffner *et al.* [47] analyzed 490 deletion-related screens across the top 20 social media platforms and revealed that while most platforms offer account deletion on desktop browsers, many lack equivalent options on mobile apps. Lingareddy *et al.* [41] analyzed the account deletion processes on the 20 most popular social media platforms and found that account deletion is often not straightforward and can be confusing for users. Santhanam *et al.* [46] developed a tool to detect and address Leftover Account Issue. They found that 64.45% of apps fail to provide users with an account deletion method and 2.5% of apps didn’t delete users’ account data after account deletion.

Distinction from existing works: Our work differs from existing work [46] in two key ways. First, our primary objective is to assess whether apps comply with Google Play’s new policy introduced in April 2023, which requires apps to provide both in-app path and web-based account deletion methods in account deletion links. Accordingly, our main contribution lies in analyzing *policy compliance* of Google Play apps. In contrast, prior work [46] (2022) focused on verifying whether account deletion is functional and whether data is retained instead of policy compliance of apps. An app may technically provide the account deletion functionality in code, which satisfies work [46], but still violate Google Play’s policy, such as missing an account deletion link. Second, in terms of methodology, our Account Deletion Verifier identifies account deletion paths using text analysis of the account deletion links. In contrast, prior work [46] identified the account deletion paths through code analysis.

8 Conclusion

In this work, we conducted the first comprehensive analysis of Google Play apps’ account deletion links and related issues. We first performed a pilot study to identify potential problems. Based on these findings, we developed a tool named DELETETRACKER to detect different account deletion issues at scale. DELETETRACKER found only 8.5% of apps to fully comply with Google Play’s account deletion requirements, while others exhibit various link and content issues. We also found that there are apps that do not delete user accounts even

after submitting account deletion requests. We have reported our findings to Google and received their acknowledgement.

Ethics Considerations

Throughout the development and application of DELETETRACKER, we prioritized adherence to ethical principles at every stage of the study and ensured the integrity of our research.

Data collection: We sourced app metadata from Andro-Zoo [28, 30], an open-source dataset. Additionally, we collected publicly accessible data, such as account deletion links, data management links from the Google Play mobile app, and the data deletion statement from the Google Play website. No user data, developer information, or sensitive personal data was accessed, stored, or used during the study.

Account management: We registered around 300 accounts in total throughout the study. Each app was assigned two registered accounts to minimize any impact on app systems. During registration, we used a virtual profile to ensure no real personal information was provided. All accounts were created using new email addresses containing no personally identifiable information. This approach ensured responsible interaction with the platforms studied.

Responsible disclosure: We have responsibly disclosed our findings and the identified account deletion issues to Google Play. Google has filed a bug based on our report and assigned it a Medium (S2) severity level, indicating that they recognized it as a significant issue. Additionally, we have notified app developers about the issues identified in their apps, encouraging them to take action.

Open Science

Our work aligns with both the ethical guidelines and open science policy of USENIX Security’25. We have made all materials used in this study publicly available, including the code, dataset, and results (<https://zenodo.org/records/15610882>).

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A Alternative Account Deletion Methods

Figure 10 shows the content of JazzCash’s account deletion link. It provides three account deletion methods: phone call, visiting a branch, and email.

You can close your JazzCash account via following methods:

- 1. Via Phone Call**
For Jazz Users: Dial 4444 from your mobile phone.
For Other Network Users: Dial 021 111 124 444
 - Follow the voice prompts to connect with a customer care representative.
 - Inform the representative of your desire to close your JazzCash account.
 - You'll be asked to verify your identity through a series of questions.
 - Upon successful verification, your account will be deactivated, and you'll receive a confirmation message.
- 2. Visiting a Jazz Franchise or Mobillink Microfinance Bank Branch**
 - Locate and visit the nearest Jazz franchise or Mobillink Microfinance Bank branch.
 - Request an account closing form from a representative.
 - Fill out the form and submit it along with a copy of your valid CNIC.
 - The representative will verify your information and proceed to close your account.
 - It's necessary to withdraw or transfer all funds from your JazzCash account before proceeding with the closure.
- 3. Via Email**
You can also request account deletion by emailing the JazzCash support team at complaints@jazzcash.com.pk. In your email, clearly state your request to close your account, providing necessary account details for identification.

Figure 10: Account deletion method provided by JazzCash.

B Account Deletion Verification Details

Table 4 presents the two virtual profiles we created for app registrations. All information provided is fabricated, ensuring that no real personal information was disclosed during the research.

Key	Value1	Value2
First Name	John	Sam
Last Name	Doe	Dorf
Username	johndoe987654321	sdfs23
Phone Number	+1-234-567-8901	+1-345-567-8901
Date of Birth	1990-01-01	2000-01-01
Address	123 Main St, Springfield	123 Main St, Springfield
Account ID	987654321	123456789
Security Question	What is your pet's name?	What is your pet's name?
Security Answer	Fluffy	Fluffy

Table 4: Virtual profiles for registration information.

C Evaluation Details

C.1 Evaluation Datasets

As shown in Table 5, we used different sets of annotated data to evaluate each module and end-to-end performance. When evaluating each module of DELETETRACKER, we assessed whether the identified issues are correct (precision) and whether any issues have been missed (recall). Since different modules are designed for detecting different types of issues, and the number of apps associated with each issue

varies, we calculated modular precision using the corresponding number of apps for each issue. To calculate recall, we randomly sampled 100 apps from 863 apps (*i.e.*, ME_{100}) to check whether any issues were missed. The details of the 100 sampled apps are shown in Table 6. After that, to evaluate the end-to-end performance of DELETETRACKER, we randomly sampled and annotated another 100 new apps (*i.e.*, EE_{100}) that were unseen in the development of DELETETRACKER.

Module	Issue	# of apps
Account Deletion Crawler	Unavailable apps	2
	Missing account deletion link	39
Account Deletion Link Analyzer	Cross-platform inconsistency	8
	Inaccessible link	5
	Missing account deletion method	13
Account Deletion Method Classifier	Alternative account deletion method	5
	Missing web-based account deletion	8
	Missing in-app path account deletion	31
	Compliance without issue	4

Table 6: Summary of issues found in 100 apps sampled from 863 apps of recall calculation.

C.2 Confusion Metric Definition for Issue-level End-to-end Evaluation

In security research (*e.g.*, vulnerability scanning or anomaly detection), a true positive (TP) occurs when a security system or tool correctly detects a vulnerability/issue. A false positive (FP) would be a system or tool incorrectly identifying a vulnerability/issue that doesn’t exist. Our definitions are consistent with the labeling criteria used for issue/vulnerability scanning tasks. The confusion metric definitions for Account Deletion Link Analyzer are shown in Table 7.

Ground Truth	DeleteTracker	Final Label
Missing link	Missing link	TP
With link	Missing link	FP
Missing link	With link	FN
With link	With link	TN
Inaccessible link	Inaccessible link	TP
Accessible link	Inaccessible link	FP
Inaccessible link	Accessible link	FN
Accessible link	Accessible link	TN
Cross-platform inconsistency	Cross-platform inconsistency	TP
Cross-platform consistency	Cross-platform inconsistency	FP
Cross-platform inconsistency	Cross-platform consistency	FN
Cross-platform consistency	Cross-platform consistency	TN

Table 7: Labeling criteria of Account Deletion Link Analyzer used to determine TP, FP, FN, and TN based on ground truth and DeleteTracker outputs.

Module	Issue	# of Apps (Precision)	# of Apps (Recall)
Account Deletion Link Crawler	-	572 apps with link	100 apps sampled from 863 apps (ME_{100})
Account Deletion Link Analyzer	Missing link	291 apps missing link	
	Cross-platform inconsistency	65 apps with cross-platform inconsistency	
	Inaccessible link	25 apps with inaccessible link	
Account Deletion Method Classifier	Non-compliant account deletion method	100 apps randomly selected from 863 apps (ME_{100})	100 new unseen apps sampled from all Google Play apps excluding the 863 apps (EE_{100})
End-to-end Evaluation	All issues	100 new unseen apps sampled from all Google Play apps excluding the 863 apps (EE_{100})	

Table 5: Number of apps used in precision and recall calculation in evaluation.

Module	Issue	TP	FP	FN	Precision	Recall
Account Deletion Link Crawler	No issue detected	-	-	-	-	-
Account Deletion Link Analyzer	Missing account deletion link	52	1	1	98.1%	98.1%
	Cross-platform inconsistency	7	1	0	87.5%	100%
	Inaccessible link	4	1	0	80%	100%
	Total	63	3	1	95.5%	98.4%
Account Deletion Classifier	Missing in-app path	26	4	8	86.7%	76.5%
	Missing web-based	22	6	1	78.6%	95.7%
	Total	48	10	9	82.8%	84.2%
End-to-end Performance	All issues	111	13	19	89.5%	91.7%

Table 8: The details of issue-level end-to-end performance.

Ground Truth	DeleteTracker	TP	FP	FN
Missing web-based	Missing web-based	1	0	0
Missing in-app path	Undetected	0	0	1
Missing in-app path	Missing method	1	1	0
Missing method	Missing web-based	1	0	1

Table 9: Examples of TP, FP, and FN calculations for account deletion method classification.

For the Account Deletion Method Classifier, since Google Play defines two account deletion methods (*i.e.*, in-app path and web-based account deletion method), we calculate precision and recall based on whether DELETETRACKER correctly detects the absence of either method. To illustrate how we compute the confusion metrics, we provide 4 examples in

Table 9. For the case of “missing account deletion method”, we treat it as two separate issues: a missing in-app path issue and a missing web-based method issue. For example, if DELETETRACKER detects only the missing web-based issue, it contributes to one TP for the web-based category (as it was correctly identified), and one FN for the in-app path category (as it was missed).

Table 8 shows how we calculate the precision and recall in issue-level end-to-end evaluation. The Account Deletion Link Crawler is not included in the issue-level end-to-end evaluation because it does not participate in the detection of any defined issues (as shown in Figure 2) when we conducted the end-to-end evaluation at the issue-level. We separately calculated this module’s precision and recall (100% and 98%). We only identified one app where the crawler failed to retrieve the account deletion link.